IEOR 173 Lecture Notes

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Chapter 1

1.1 Lecture 03

1.1.1 Joint Distributions

$$P(x, y) = P(X \leq x, y \leq y)$$

This is a cumulative distribution for joint distributions.

If $X \perp Y$ (independent), (perpendicular), thee the join distribution is the product of the marginals. $P(X, Y) = P(X)P(Y) \forall X, Y$

1.1.2 Cov + Correlations

Covariance is a measure of linear dependence.

$$Cov(X, Y) = E[(x - EX)(Y - EY)] = EXY - EXEY$$

Interpretation: If they vary together, Cov is larger.

Properties

1. Cov(X,X) = Var(x)
2. Cov(X,Y) = Cov(Y,X)
3. Cov(cX, Y) = cCov(X,Y)
4. Cov(X, Y+z) = Cov(X,Y) + Cov(X,Z)

Correlation:

$$p(X, Y) = \frac{Cov(X, Y)}{sd(X) \cdot sd(Y)}$$

If

- ¿ 0, pos correlation
- ¡ 0, negative correlation
- = 0 No correlation

Note:-

Independent does imply uncorrelated, however uncorrelated does not imply independent.
Ex: If $P(-1, 1) = P(0, 0) = P(1, 1) = \frac{q}{3}$. then $E[XY] = \frac{0}{3} = 0$. However, $E[X] = 0, EY =?$, therefore $E[XY] = 0$ and $Cov(X, Y) = 0$. There is no linear relationship, however there is a dependency for $y = x^2$.

Therefore, we can generalize our variance formula

$$Var(A + \sum b_i X_i) = \sum b_i^2 Var(X_i) + \sum_{\forall ij, i \neq j} b_i b_j Cov(X_i, X_j)$$

1.1.3 Hat Example

Example 1.1.1 (Lists of number)

Imagine go to party, men take off hats, women keep on hats. Each man randomly takes a hat and leaves.
Let $X = \#$ of matches in N pairs (man and his hat)
What is EX and Var(X)?
If two men, the only possibility of matches are 0 and 2. The mean is 1. If three men, a similar pattern occurs.

$$EX = 1 \forall N$$

As N grows larger, the probability decreases, but there is more men to balance it out.

$$Var(X) = 1 \forall N \geq 2$$

We can prove this by using Indicators.
Let $X = \sum^N I_i$, where $I_i = I$it is a match $\sim Bern(\frac{1}{N})$

$$EX = E[\sum I] = \sum^N E[I] = N\frac{1}{N} = 1$$

$$Var(I_i) = E[I^2] - [EI]^2 = p - p^2 = p(1-p) = \frac{1}{N} - \frac{1}{N^2} = \frac{N-1}{N^2}$$

$$Cov(I_i, I_j) = E[I_i I_j] = EI_i EI_j$$

$$E[I_i I_j] = P(\text{both i and j match}) = P(I - i = I_j = 1)$$

$$P(I_i = 1)P(I_j = 1 \mid I_i = 1) = \frac{1}{N} \cdot \frac{1}{N-1}$$

Conceptually, we can think of event j following event j

$$Cov(I_i, I_j) = \frac{1}{n(n-1)} - \left(\frac{1}{n}\right)^2 = \frac{1}{N^2(N-1)}$$

$$Var(X) = Var(\sum I_i) = \sum Var(I) + \sum Cov(I_i, I_j) = N \cdot \frac{N+1}{N^2} + N(N-1)\frac{1}{N^2(N-2)} = 1$$

The calculation will be left as an exercise to the future reader.
As $N \to \infty$,

$$Cov(I_i, I_j) \to 0$$

$$Var(I_i) \to \frac{1}{N}$$

$$X \to Bin(N, \frac{1}{N}) \sim Poisson(1)$$

Intuitively, if there are thousands of men there, whether or not 1 person gets his hat is not really going to affect the chances another gets his.

1.1.4 Start condition (Ch 3)

What if we have dependent random variables and want to compute conditional probabilities?

Conditions:

1. discrete: $P(X, y) = P(X = x, X = y), P(x, y) = P(X \leq x, Y \leq y) = \sum_{\leq x} \sum_{\leq y} P(x, y)$
2. continuous: $f(x, y) = F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v) du dv$

3. marginal: $F_X(x) = P(X \leq x) = F(X, \infty)$
4. conditional: $P_{x|y}(X \mid y) = P(X = x|Y = y) = \frac{P(X=x,Y=y}{P(Y=y)} = \frac{P_{x,y}(X,y)}{P_y(y)} = \frac{\text{joint}}{\text{marginal}}$

Chapter 2

2.1 Lecture 04

2.1.1 Properties of condition expectation

$$E[X|X = x] = x \forall x$$

$$E[X|Y = y]$$

, is a number bu $E[X|Y]$ is a random variable.

$$X \perp W, X \sim exp(\lambda), W \sim exp(\lambda)$$

$$f_x(x) = \lambda e^{-x} \text{for x} \geq 0$$

$Y = X + W$, condition on $Y = y$, $E[X|Y = y]$. We need $f_{x|y}(x|y)$

$$X|y + W = [y - X|y]$$

This should have the same distributions.

To show this, we need to get the distribution function.

$$f_{x|y}(x|y)\frac{f_{x,y}(x,y)}{f_y(y)} = \frac{f_{x,w}(x, y - x}{f_y(y)} = \frac{f_x(x)f_w(y - x)}{f_y(y)} = \lambda e^{-\lambda x}\lambda e^{-\lambda(y-x)} = \lambda^2 e^{-\lambda y}$$

For $0 \leq x \leq y$

We also need to get the marginal distributions.

$$f_y(y) = \lambda^2 \int_{-\infty}^{\infty} f_{x,y}(x,y)dx = \lambda^2 \int_0^y e^{-\lambda y}dx = \lambda^2 y e^{-\lambda y}$$

$$f_{x|y}(x|y) = \frac{f_{x,y}(x,y)}{f_y(y)}\frac{\lambda^2 e^{-\lambda y}}{y\lambda^2 e^{\lambda y}} = \frac{1}{y}$$

This is a uniform distribution from $0 \leq x \leq y$. Therefore $E[X|Y = y] = \int_0^y (x\frac{1}{y})dx = \frac{y}{2}$

Instead, if we take $G(y) = E[X|Y]$, which is a random variable, we can get $E[E[X|Y]$.

We can do this by $E[g(y)] = \int_{-\infty}^{\infty} g(x)D(x)dx$, which is the probability $g(X)$ multiplied by the density $d(x)$.

$$E[E[X|Y]] = \int E[X|Y = y] f_y(y) dy$$

.

It turns out that this becomes $E[X]$.

Example 2.1.1
$Y = \sum^N X_i, EX_i = \mu$, $N \perp X_i$'s, or N is independent of the X_is.

$$E[Y] = E[\sum^N y_n] = E[E[\sum^N X_i|N]] = \sum_{\text{all } n} E[\sum^N X_i|N = n] p_n(n) = \sum_{\text{all} n} E[\sum m^n X_1] p_n(n) = \sum_{\text{all} n} n\mu p_n(n)$$

$$= \mu \sum np_n(n) = \mu EN$$

Where N is discrete, integer valued, nonnegative. $p_n(n)$ is the pmf for n

Example 2.1.2 (Mean of a geometric)

$$X \sim geom(p)$$

Find EX.
Because geometric dists are memoryless, we can condition on the first step.
Let $I = I\{\text{1st trial is a success}\} \sim Bern(p)$
Compute EX

$$EX = E[E[X|I]] = \sum_{\text{all i}} E[X|I = i] p(I = i) = E[X|I = 1] p(I = 1) + E[X|I = 0] p(I = 0)$$

$$= 1p + (1 + EX)(1 - p) \implies EX = \frac{1}{p}$$

Basically saying, we either get the heads on this trial, or then we have to restart.

Example 2.1.3 (Packet Transmission)
Serve 1 packet per time slot. Each slot we have $A_i = \#$ arrivals in slot i. We assume i.i.d. and has p_0, p_1, p_2.
We assume that the buffer is size 2. (We can only hold 2 packets in the buffer)
Basically, for every time slot, we can only store up to 2 packets. Each timeslot, one packet leaves the buffer, and if there is too many packets, the incoming packet has to bounce.

$$N = \text{length of busy period}$$

A busy is period is where there are arrivals to an empty system and ends when the system is emptied again.
Need $E[N]$
First thing that we need to think about is conditioning. It would be helpful to know how many packets start the busy period, therefore we condition on $X = \#$packets that start the busy period (b.p), where $X \in \{1, 2\}$. Thus we also need to know $P(X = 1) = P(A = 1|A = 1\text{or} = 2) = \frac{p_1}{p_1+p_2}$ and $P(X = 2)i = P(A = 2|\dots) = \frac{p_2}{p_1+p_2}$.

$$E[N] = E[E[N|X]] = EN|X = 1]\frac{p_1}{p_1 + p_2} + E[N|X = 2]\frac{p_2}{p_1 + p_2}$$

Chapter 3

3.1 Lecture 05

3.1.1 Packet Transmission (cont)

Example 3.1.1 (Packet Transmission (cont))

Packet transmission appends A_i i.i.d. p_0, p_1, p_2. N = length of busy period. We still want $E[N]$

$$X = \begin{cases} 1 \frac{p_1}{p_1+p_2} \\ 2 \frac{p_2}{p_1+p_2} \end{cases}$$

$$E[N] = E[E[N|X]] = E[N|X=1]\frac{p_1}{p_1+p_2} + E[N|X=2]\frac{p_2}{p_1+p_2}$$

We now need to compute $E[N|X=n]$.
For simplicity $E[N_1] = E[N|X=1]$:
It would help to know how many packets we get at time 1

$$E[N_1] = E[E[N_1|A]]$$

where A is the number of arrivals in the slot of the busy period.

$$\begin{aligned} E[N] &= E[N_1|A=0]p_0 + E[N_1|A=1]p_1 + E[N_1|A=2]p_2 \\ &= 1p_0 + (1+E[N_1])p_1 + (1+E[N_2])p_2 \\ &. \end{aligned}$$

We can use the $1 + E[N_1]$ because of the i.i.d. and the memory less state.

$$E[N_2] = E[N_2|A=0]p_0 + E[N_2|A=1]p_1 + E[N_2|A=2]p_2$$

$$E[N_2] = (1+E[N_1])p_0 + (1+E[N_2])p_1 + (1+E[N_2])p_2$$

The last term is because the buffer only has storage for 2 packets. If one goes out, then one gets replaced and the other gets sent away.
We can thus solve for $E[N_1]$ and $E[N_2]$, which we can plug into the original equation.
This is a very simple Markov Chain.

3.1.2 Computing Variance:

$\mathrm{Var}(x)$ in two methods

1. Use $\text{Var}(x) = E[X^2] - E[X]^2$, condition for $E[X^2] + E[X]$
2. Condition Var formula

   Example for method 1:

$$X \sim geo(p)$$

$$E[X^2] = E[E[X^2]|I] \text{where} I = \text{indicator of success } \sim \text{Bern}(p)$$

$$\begin{aligned} E[X^2] &= E[X^2|I = 1]p + E[X^2|I = 0](1-p) \\ &= 1p + (E[(X+1)^2])(1-p) = p + E[X^2 + 2x + 1](1-p) \\ &= p + (E[X^2] + E[2X] + 1)(1-p) \end{aligned}$$

   Solve for $E[X^2] = \frac{n-p}{p^2}$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2}$$

   How do we get the method 2 formula?

Proof. Conditional Variance Formula

$$\begin{aligned} [E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]) &= E[E[X|Y^2] - (E[X|Y])^2] + E[E[X|Y]^2] - E[E[E[X|Y]^2]]^2 \\ &= E[E[X^2|Y]] - E[E[X|Y]]^2 + E[X^2] - E[X]^2 \\ &= \text{Var}(X) \end{aligned}$$

.

Theorem 3.1.1 $X \perp Y \implies \text{Var}(XY) = \text{Var}(X)$

$$X \perp Y \implies \text{Var}(XY) = E[Y^2]\,\text{Var}(X) + E[X]^2\,\text{Var}(Y) = E[X^2]\,\text{Var}(Y) + E[Y]^2\,\text{Var}(X)$$

$$\text{Var}(XY) = E[\text{Var}(XY|Y)] + \text{Var}(E[XY|Y])$$

.

For the first term:

$$\text{Var}(XY|Y = y) = \text{Var}(Xy|Y = y) = y^2\,\text{Var}(X) \implies \text{Var}(XY|Y) = Y^2\,\text{Var}(X)$$

We can do this because X is independent of Y.

For the second term:

$$E[XY|Y = y] = yE[X] \implies E[XY|Y] = YE[X]$$

Therefore,

$$\text{Var}(XY) = E[Y^2\,\text{Var}(X)] + \text{Var}(YE[X]) = \text{Var}(X)E[Y^2] + E[X]^2\,\text{Var}(Y)$$

Theorem 3.1.2

X_i is i.i.d. N is an integer valued random variable independent of the X_i s. Find the $\mathrm{Var}(\sum^N X_i)$

$$\mathrm{Var}(\sum^N X_i) = E[\mathrm{Var}(\sum^N X_i|N)] + \mathrm{Var}(E[\sum^N X_i|N)]$$

For the first term:

$$\mathrm{Var}(\sum^N X_i|N = n) = \mathrm{Var}(\sum^n X_n|N = n) = \sum^N \mathrm{Var}(X|N = n) = \sum^n \mathrm{Var}(X) = n\,\mathrm{Var}(X_i)$$

We proved the second term earlier.

$$E[\mathrm{Var}(\sum^N X_i)] = E[N\,\mathrm{Var}(X_i)] + \mathrm{Var}(NE[X_i]) = E[N]\,\mathrm{Var}(X_i) + \mathrm{Var}(N)E[X_i]^2$$

If $N \perp X$

$$\mathrm{Var}(NX) = E[X^2]\,\mathrm{Var}(N) + E[N^2]\,\mathrm{Var}(X)$$

This one is like the catering example, where everyone orders the same thing. The other is if all the X_i are i.i.d., therefore ordering different things. Therefore the first is more variable than the second.

Chapter 4

4.1 Lecture 06

4.1.1 Recall:

$$I\{A\} \sim \text{Bern}(P(A)) \implies E[I\{A\}] = P(A)$$

$$EX = E[E[X \mid y]]$$

Therefore,

$$\begin{aligned}
P(A) &= E[I\{A\}] \\
&= E[E[I\{A \mid Y\}]] \\
&= E[P(A \mid y)] \\
&= \sum_y P(A \mid Y = y) p_y(y) \quad \text{or} \int_{-\infty}^{\infty} P(A \mid Y = y) f_y(y) dy \\
&= \text{Total probability rule}
\end{aligned}$$

These concepts are important for Markov chains!

Example 4.1.1

i.i.d. rolls of 2 fair dice. What the chance of sum =5 before sum =7
We could enumerate all the possibilities.

$$P(\text{5on a single trial}) = \frac{4}{36}$$

$$P(\text{7on a single trial} = \frac{6}{36}$$

We can do the computation 2 ways.

$$P(5 \text{ before } 7) = P(E)$$

Conditioning!!

Let X = sum on 1st roll.

$$\begin{aligned}
P(E) = E[P(E|X)] &= P(E \mid X = 5)P(X = 5) + P(E \mid X = 7)P(X = 7) + P(E \mid X = \text{other}P(\text{other})) \\
&= 1 \cdot \frac{4}{36} + 0\frac{6}{36} + P(E)\frac{26}{36} \\
&= \frac{4}{10} = \frac{2}{5}
\end{aligned}$$

Another way to do it:

$$P(E) = E[P(E \mid N)]$$

Where N is the number of rolls until 5 or 7. $N \sim geom(\frac{10}{36})$

$$\begin{aligned}
P(E) &= \sum_{n=1}^{\infty} P(\text{sum to 5 on nth roll for 1st time} \mid N = \text{first 5 or 7})P(N = n) \\
&= \sum_{n=1}^{\infty} P(\text{sum to 5 on roll n} \mid \text{sum to 5 or 7 on roll n })P(N = n) \\
&= \sum_{n=1}^{\infty} \frac{\frac{4}{36}}{\frac{4}{36} + \frac{6}{36}} P(N = n) \\
&= \sum_{n=1}^{\infty} \frac{4}{10} P(N = n) \\
&= \frac{2}{5} \sum_{n} P(N = n) \\
&= \frac{2}{5} \cdot 1
\end{aligned}$$

because of i.i.d., it doesn't matter whether if its the first time or not
We show here that $E \perp N$.

4.1.2 Markov chains

A stochastic process is a sequence of random variables.

$$\{X_n\} = \text{ stochastic process } = \text{ sequence of R.Vs}$$

Simplest sequence: X_n i.i.d. This has no memory and is stateless.

A Markov chain only considers the most recent thing that happened

MC = markov chain, DTMC = discrete time markov chain.

Definition 4.1.1: Discrete Time Markov Chain

A markov chain is defined as

$$X_{n+1} \mid X_n, X_{n-1}, \dots X_0 \perp X_{n-1}, \dots, X_n$$
$$P(X_{n+1} \mid X_n =, X_{n-1} =, \dots) = P(X_{n+1}|X_n) = P_{ij}$$

In other words, only the previous state matters.
For us, time homogeneous, N doesn't matter

$$P = \|P_{ij}\|$$

This is a transition matrix, where all elements are nonnegative, and the row sum = 1. From state i to state j.

Example 4.1.2 (Gilbert Model of transmission)

G is good State, B is Bad State
We could either stay in each node, or to move between good and bad.
Transition Matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

Example 4.1.3 (4 state markov chain:)

$$X \in \{GG, GB, BG, BB\}$$

$$P(G \mid X) = p_1, p_2, p_3, p_4$$

Going from GG to GG, the probability is p_1. going from GG to GB, that would be $1 - p_1$.
From GB to BG, the probability is p_2. Going from GB to BB, the probability $1 - p_2$
etc.

Example 4.1.4 (Birth Death Markov Chain)

Infinite number of states, but only 2 transmissions. You can either go up and down, or stay in the same state. see textbook for more.

Example 4.1.5 (Inventory Model)

D_i i.i.d. daily demand. If inventory $\leq s$ at day end, order up to S to start next day. (s, S) policy.
State = start of day inventory level

States are from $\{s+1, s+2, i, S-1, S\}$
To stay at state i , $P(D_i = 0)$. to up to S, $P(D_i \geq i - s)$. To go down to $i - 1$, $P(D_i = 1)$, etc.

Chapter 5

5.1 Lecture X: 25 February 2025

$\{X\}$ irreducible $\iff$ there is only one class for all states $(i \leftrightarrow j \forall i, j \in S$

Definition 5.1.1: Period

$$d(i) = \gcd\{N \geq 1 : P_{ii}^n > 0\}$$

i has period $d = d(i)$:

$$P_{ii}^n > 0 \implies \text{d divides n}$$

And all states have the same period.

n is not a multiple of $d \implies P_{ii}^n = 0$, thus aperiodic : $d(i) = 1$

Example 5.1.1 (Period of 2)

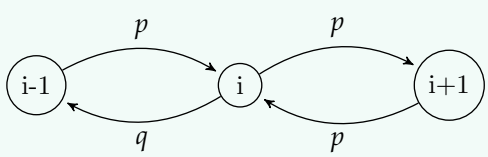


Figure 5.1: period of 2

there are more nodes past $i+1$ with same structure in a birth death cycle. Therefore, you can only get to even nodes or odd nodes depending on i.

5.1.1 Stationary Distribution

X: $X_n \sim \pi$, where π is a row vector for all the probabilities. π is the stationary distribution (equilibrium/invariant) n, if $X_n \sim \pi \implies X_{n+1} \sim \pi$

Example 5.1.2 (Gilbert Model)

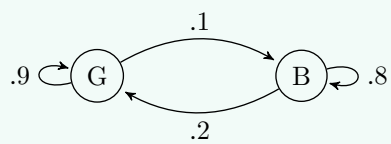


Figure 5.2: Gilbert model

Our intuition states that the stationary dist n should be $\pi = (\frac{2}{3}, \frac{1}{3})$, because we are staying i. $\frac{.2}{.3}$ of the time into G.

$$\pi_G = P(\text{end up in G})) = P(\text{start in G and stay or start in B and move})$$

$$= \pi_g(1-p) + \pi_B(q)$$
$$= p\pi_g = q\pi_B$$

If we solve this for B as well, and use the fact that $\pi = 1$. We get $\pi = (\pi_g, \pi_b) = (\frac{q}{p+q}, \frac{p}{p+q})$
Plugging in $p = .1, q = .2$, we get our original answer.

$$\pi_G = \text{rate of transitions into or out of G}$$

In other words, detailed balance of $\pi_G(1-p) = \pi_B q$

Example 5.1.3 (Packet Transmissions)
Packet example: (G/D/1/2) (Incoming, Outgoing (Deterministic), server #, memory)

$$A_i \sim (p_0, p_1, p_2)$$

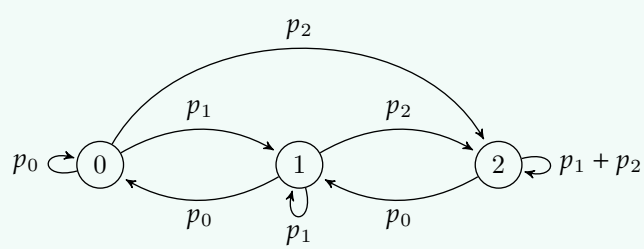


Figure 5.3: packet transmission

What is the stationary dist n?

$$\pi_0 = \pi_0 p_0 + \pi_1 p_0 + \pi_2 \cdot 0$$
$$\pi_1 = \pi_0 p_1 + \pi_1 p_1 + \pi_2 p_0$$
$$\pi_2 = \pi_0 p_2 + p_1 p_2 + \pi_2(p_1 + p_2)$$
$$\pi_1 + \pi_2 + \pi_3 = 1$$

This results in $\pi = (\frac{p_0}{1-p_1} + \frac{p_0(1-p_0)}{1-p_0}, \frac{p_2}{1-p_1})$

Theorem 5.1.1 Stationary Distribution Computation

If π satisfies

1. $\pi_j = \sum_{i \in S} \pi_i P_{ij} \forall j \iff \pi = \pi P$
2. $\sum_{i \in S} \pi_1 = 1$

Then π is a stationary distribution

Proof. Suppose $X_0 \sim \pi$, We need to show that $X_1 \sim \pi$

For $P\{X_1 = j\} = \pi_j$

$$P\{X_1 = j\} = E[P(X_{1=j|X_0})] = \sum_{i \in S} P(X_1 = j \mid X_0 = i) P(X_0 = i) = \sum_{i \in S}^{P_{ij} \cdot \pi_i}$$

Definition 5.1.2: Steady State

The steady state distribuion of $\{X\}$, when it exists, is defined for all j by

$$\lim_{n \to \infty} P(X_n = j \mid X_0 = i) = \lim_{n \to \infty} P_{ij}^n = \text{steady state prob of j}$$

Definition 5.1.3: Long run time average distribution

the long run time distribution of $\{X\}$, if it exists, is defined for all j by

$$\lim_{n \to \infty} \frac{1}{n} \sum_{m=0}^{n} I\{X_m = j \mid X_0 = i\} = \text{l.r. prop time in j}$$

This is weaker than the steady state, basically like an average. counting the number of times we are in state n, resulting in the indicators.

Theorem 5.1.2

For irreducible periodic, positive recurrent MC,

$$P_{ij}^n \to \pi_j, \quad \text{where } \pi \text{ is the stationary distribution}$$

Theorem 5.1.3

For irreducible positive recurrent MC, long run proportion time in j $= \pi_j$, where π is the stationary distribution.

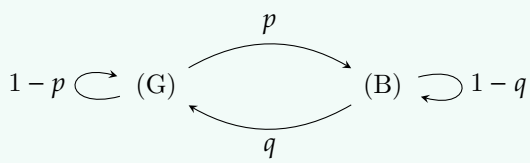
a

Chapter 6

6.1 Lecture 09

6.1.1 Stationary Distribution

Example 6.1.1 (Gilbert Model)



For the gilbert Model: If $p = .1$ and $q = .2$

$$\pi_G = \pi_B(q) + \pi_G(1-p)$$

$$\pi_B = \pi_B(1-q) + \pi_G p$$

AND

$$\pi_G + \pi_B = 1$$

$$\pi = (\pi_G, \pi_B) = (\frac{q}{p+q}, \frac{p}{p+q})$$

$$\begin{aligned} P_{11}^1 &= .9 \\ P_{11}^2 &= .9^2 + (.1)(.2) = .82 \\ P_{11}^3 &= .781 \\ &\vdots \\ P_{11}^n &= \frac{2}{3} \end{aligned}$$

This converges because the chain is aperiodic, irreducible, finite state.

Example 6.1.2 (Packet Transmission Revisited)

For the Packet Transmission 5.3: G/D/1/2

$$A_i \sim (p_0, p_1, p_2)$$

if $p_0 = p_1 = \frac{1}{4}$ We will get $\pi = \left(\frac{1}{12}, \frac{1}{4}, \frac{2}{3}\right)$
The $P(\text{busy})$ in the long run, it will be $\frac{11}{12}$.

What is the loss rate? $E[\#$ of packets lost/slot$]$

$$\begin{aligned} E[\# \text{ of packets lost}] &= \pi_2 p_2(1) \\ &= \frac{2}{3} \cdot \frac{1}{2} \\ &= \frac{1}{3} \end{aligned}$$

On average, once every 3 time slots, we lose a packet.

What proportion of packets are lost? We need the arrival rate = # packets coming per slot.

$$\begin{aligned} E[A_i] &= 0p_0 + 1p_1 + 2p_2 \\ &= \frac{1}{4} + 2 \cdot \frac{1}{2} = \frac{5}{4} \end{aligned}$$

Thus $\frac{1}{3} \div \frac{5}{4} = \frac{4}{15}$. This is the number that are lost per slot divided by number of packets arriving per slot. Thus, about .272 of the packets are lost.

What is the effective service rate? What is the through put of the system?

$$\begin{aligned} \text{throughput} &= \text{proportion served} \times \text{arrival rate} \\ &= (1 - \text{prop lost}) \cdot \text{arrival rate} \\ &= (1 - \frac{4}{15}) \cdot \frac{5}{4} \\ &= \frac{11}{12} \\ &= P(\text{busy}) \cdot 1\text{served per slot} \end{aligned}$$

Example 6.1.3 (Google Search Rankings)

What order to put pages in? Develop a giant markov chain, with the websites being a state and the probability to visit that state.

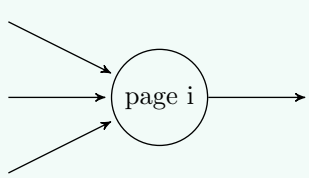


Figure 6.1: Google Page Ranking

So multiple sites pointing into page i, and then also they have a weighted arrows going out.

6.1.2 Special Cases

p is doubly stochastic.

For a doubly stochastic markov chain, the rows and the columns sum up to 1.

In this case, all the states are equally likely, or a uniform distribution

$$\pi = (\frac{1}{n}, \frac{1}{n}, \dots$$

Cut sets: ∀cut If we partition our states, the rate across one way is equal to the rate across the other way . This is especially helpful in birth death processes (ex. Gilbert Model)

If we cut the Gilbert Model down the middle,

$$\pi_G p = \pi_B q$$

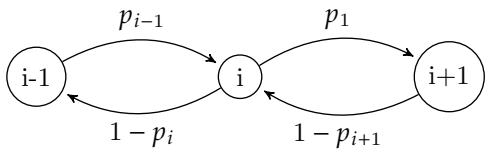


Figure 6.2: Birth Death

If we cut this between i and i+1,

$$\begin{aligned}
\pi &= \pi P \\
\pi_i &= \pi_{i-1} p_{i-1} + p i_{i+1} p_{i+1} \\
\pi_{i+1} &= \pi_i (\frac{p_1}{p_{i+1}})
\end{aligned}$$

$$m_{jj} = E[\text{\# of transitions, back to j}] = E[T_j]$$

Thm: For an irreducible positive recurrent MC, $\pi_j = \frac{1}{E[T_j]} = \frac{1}{m_{jj}} \forall j$

For the gilbert Model, $p = .1, q = .2$. If we want the $E[T_g]$ or the time to get from good to good.

$$\begin{aligned}
E[T_g] &= \frac{1}{\pi_g} \\
&= \frac{1}{\frac{2}{3}} \\
&= \frac{3}{2}
\end{aligned}$$

The long way:

$$\begin{aligned}
E[T_g] &= .9(1 + 0) + .1(1 + E[T_{BG}]) \\
&= 1 + .1E[T_{BG}] \\
&= 1 + .1(5) \\
&= \frac{3}{2}
\end{aligned}$$

T_{BG} has a geometric distribution, $\sim geometric(.2)$. Therefore $E[T_{BG}] = 5$

Chapter 7

7.1 Lecture 11: 4 March 2026

Suppose $\{X_n\}$ has stationary distribuion π.

- Let X bc r.v $X \sim \pi$
- Let r be a function $S \to \mathbb{R}$ s.t. $r(x)$ be the cost when state is x.

Definition 7.1.1: LRAC

The long run average cost (LRAC) is:

$$LRAC = \lim_{n\to\infty} \sum_{n=0}^{N} \frac{r(x_n}{N}$$

Theorem 7.1.1 When LRAC, exists, LRAC $= E[r(x)]$

$$\begin{aligned} LRAC &= E[r(x)] \\ &= \sum_{i\in S} r(i)\pi_i \end{aligned}$$

A sufficient condition for LRAC existence:

- $\{X\}$ irreducible, positive recurrent and r bounded or linear.

Corollary 7.1.1 When $r(x)$ is linear, $r(x) = a + bx$ then LRAC $= a + bE[X], X \sim \pi$

Corollary 7.1.2 If you pay \$1 per unit time in state j, 0 otherwise. Then $r(x) = I\{X = j\}$ and LRAC $= \pi_j$

Example 7.1.1

Inventory: $s = 0, S = 3$, 0 lead time, no back orders. X_n = inventory at start of day. Daily demand D_i i.i.d. and $D_i \sim (.2, .4, .1, .3)$

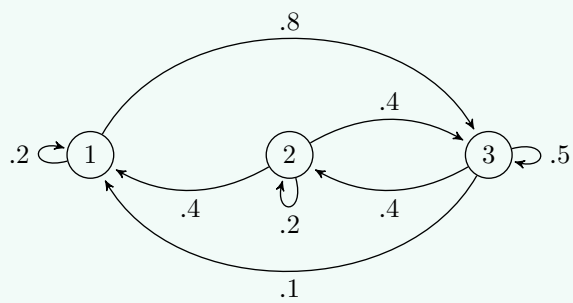


Figure 7.1: Inventory Model

Note:-

Because we only order when we get to 0 and only order at the end of the day, we can only go from $1 \longrightarrow 1$ and $1 \longrightarrow 3$. Similarity for 3. We can get to 3 with $D_i = 0$ or $D_i = 3$ because we can order at the end of the day.

$$\mathbb{P} = \begin{bmatrix} .2 & 0 & .3 \\ .4 & .2 & .4 \\ .1 & .4 & .5 \end{bmatrix}$$

We can solve the balance equations $\pi = \pi P$.

$$\begin{aligned} \pi_1 &= .2\pi_1 + .4\pi_2 + .1\pi_3 \\ &\vdots \\ \pi &= 1 \end{aligned}$$

$$\pi = \left(\frac{1}{5}, \frac{4}{15}, \frac{8}{15}\right)$$

$$\begin{aligned} E[X] &= 1 \cdot \frac{1}{5} + 2 \cdot \frac{4}{15} + 3 \cdot \frac{8}{15} \\ &= \frac{7}{3} \\ &= 2.33 \end{aligned}$$

So the long run average holding cost if costs is \$2 unit/day:

$$\begin{aligned} E[r(x)] &= 2 \cdot 1\pi_1 + 2 \cdot 2\pi_2 + 2 \cdot 3\pi_3 \\ &= 2E[X] \\ &= 4.67 \end{aligned}$$

So on average, we are paying \$4.67 per day in holding costs.

Example 7.1.2 (Extension of 7.1.1 with Inventory Model)

What is LRAC w/ holding cost (h.c.) \$2 per unit per day, ordering cost \$3 per order + stockout cost = \$4 unit stocked out.

For ordering cost, need reorder rate (RR) = long run rate of transitions triggering an order. Or in other words, long run proportion of days where demand is ≥ supply .

$$\begin{aligned} RR &= \pi_1(P(D_i \geq 1) + \pi_2(P(D_i \geq 2)) + \pi_3(P(D_i \geq 3)) \\ &= .8 \cdot \frac{1}{5} + .4 \cdot \frac{4}{15} + .3 \cdot \frac{8}{15} \\ &\approx .43 \end{aligned}$$

So the l.r average reorder cost is \$3 per order(.43 orders per day) = \$1.29 per day.

Stock out cost:
We need long run average number of units stocked out. (SR) j

$$\begin{aligned} SR &= \pi_1(1P(D_i = 2) + 2P(D_i = 3)) + \pi_2\,(2P(D_i = 3)) \\ &= \pi_1(.1 + 2 \cdot .3) + \pi_2(.3) \\ &= .22 \end{aligned}$$

We have on average 1 stock out every 4 days. Therefore LR average stock out cost is (\$4 per unit)(.22 units/day) = \$ .88 per day.

Now we have the total costs, we can start the analyze the policy and optimize for min costs.

Example 7.1.3 ($S_{ij} = P(\text{ever get to j} \mid X_0 = i)$)

$S_{ij} = P(\text{ever get to j} \mid X_0 = i)$

If j is recurrent and $j \in R$. i is also recurrent.

If i, j is in the same recurrent class, then $f_{ij} = 1$
If $i \in R' \neq R$, then $f_{ij} = 0$, because recurrent classes are closed.

If i transient and j recurrent , or j transient then we change to make absorbing so then it makes it recurrent.

$$T = \text{set of transient states}$$

Let $j \in R \forall i \in T$:
Then

$$f_{ij} = \sum_{k \in T} P_{ik} f_{kj} + \sum_{k \in R}^{P_{ik}} 0$$

Basically, if we go to a node in T, or go to the rccurent state, or go into another reccurent class.

Chapter 8

8.0.1 Lecture 12: 9 March 2026

Chapter 9

9.1 Lecture 13: 11 March 2026

Exponential Distribution

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

λ is the rate, and $SD(X) = \frac{1}{\lambda}$ and $\text{Var}(X) = \frac{1}{\lambda^2}$ Property:

1. $X \sim exp(\lambda) \implies X$ memoryless (see Data 140 notes) k
2. $P(X_1 < X_2) = P(\min(X_1, X_2)) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$

   Proof.

$$\begin{aligned} P(X_1 < X_2) &= E[P(X_1 < X_2 \mid X_1)] \\ &= \int_0^\infty P(X_1 < X_2 \mid X = x)\lambda e^{-\lambda x} dx \\ &= \int_0^\infty P(X_2 > X_1)\lambda_1 e^{-\lambda_1 x} dx \\ &= \int_0^\infty e^{-\lambda_2 x}\lambda_1 e^{-\lambda_1 x} dx \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \end{aligned}$$

$\min(X_i) = exp(\sum \lambda_i$

Proof. Show that $P(\min(X_1) > x) \leq e^{-\sum \lambda x}$. This is a tail distribution.

$$\begin{aligned} P(\min(X_i \geq x) &= \prod P(X_i \geq x) \\ &= \prod e^{-\lambda_i x} \\ &= e^{-\sum \lambda_i x} \end{aligned}$$

Thus, $EX_t = EX = \frac{1}{\lambda}$, while $E[X \mid X > t] = t + \frac{1}{\lambda}$

Example 9.1.1 (Post Office)

Post office serves based on $exp(\lambda)$. Brown, Smith and Jones walk in, and they are served i.i.d.. What is the probability that Smith is the last to leave?

Jones and Brown are initally being served.
Suppose that Jones finishes first, and Smith comes in. They are actually equally likely to leave. $\frac{1}{2}$

What is the probability that Jones is the last to leave?
Then it would be $\frac{1}{4}$ by the memoryless property.

Example 9.1.2 (Failure rates)

if $X_1 \sim ex(1), X_2 \sim ex(2)$
then $P(X_1 < X_2) = \frac{1}{3}$ and the $\min(X_1, X_2) \sim exp(3)$

Chapter 10

MT 2

10.1 8 April 2026

Properties of the Poisson Process

- Stationary increments
- Independence:
- N(t) $\sim$ Poisson(λt)
- N(h) $\sim$ Bern(λh), $h \to 0$
- $T_i exp(\lambda)$, i.i.d.

10.1.1 Nonhomogenous Poisson Processes (NHPP)

Given $NHPP(\lambda \mid t)$, we relax stationary

- $m(t) = \int_0^t \lambda(s)ds$
- Counting Process

10.1.2 Complex Poisson Processes (CPP)

$\{XH\} \sim$ CPP(λ, F), $X(t) = \sum_{i=1}^{N} Y_i$, where $Y_i \sim F$ i.i.d.

Thus, $\{NH\} \sim PP(\lambda)$

We have stationary increments because we have batch sizes. We also have independent increments, as the Y_i are independent.

We can still split and join for our complex poisson process. Split is easy, but join is difficult because we have to think about the speed of each Poisson.

Join = $\{X(t)\} \sim CPP(\lambda_i, F_i)$.

So like insurance claims based on different types of claim, where F_i is size of claims.

$$\text{Multiple}\{X(t)\} \to \{X(t)\} \sim CPP(\sum \lambda_i, F)$$

Where $F = F(x) = \sum_i \underbrace{\frac{\lambda_i}{\sum \lambda_j}}_{\text{which event}} F_i(x)$

Example 10.1.1 (Properties of Complex Poisson Process)
$\{X_1(t)\} \sim CPP(1, uniform(2,4))$. $\{X_2(t) \sim CPP(2, uniform(3,4))\}$
A process comes from process 1 with probability $\frac{1}{3}$, therefore the density from 2-3 is $\frac{1}{2}$
Between 3 to 4, it will be $\frac{5}{6}$. This is because coming from X_2 is 1 with probability $\frac{2}{3}$, and from X_1 of $\frac{1}{2}$ $\frac{1}{3}$
$\frac{1}{2} \cdot \frac{1}{2} + \frac{2}{3} \cdot 1 = \frac{5}{6}$

10.1.3 Mixed Poisson Process

Let $\{N(t)\} = MPP(G)$, $\{N(t) \mid \Lambda = \lambda\} \sim PP(\lambda)$, $\Lambda \sim G$

This is if we don't know what the arrival rate is, and thus G is a distribution of what λ rate could be.

It still has stationary increments.

It does not have independent increments, because if we know the num arrivals in 1 hour, then that might change our opinion about what happens in the next hour. There is a sense of learning happening here. (Bayes Law)

10.1.4 Markov Modulated Poisson Process

This says what if λ is changing based on its own process. Could be its own DTMC or CTMC.....

For example, if the economy is good, the rate might be higher, or when its bad it would be lower. But we don't know when this would change.

There are also Renewal Processes, and Brownian Motion.

10.1.5 Continuous Time Markov Chain (CTMC)

We still assume continuous time, but discrete space.

Definition 10.1.1: CTMC Definition 1

$\{X(t)\}$ is a CTMC if it satisifies the Markov property, which is the future is independent of the past, given the present.

Mathematically, $P\{X(t+s) = i \mid X(n), 0 \le n \le s\} = P\{X(t = s) = i \mid X(s)\}$ It is the unique process with cont time and discrete space, but still satifies the Markov Process.

Definition 10.1.2: CTMC Definition 2

$\{X(t)\}$ is a CTMC if it changes states according to an embedded DTMC, with transition matrix $\mathbb{P}$ and when in state i changes after exp time with rate V_i
For now, we can assume that P has no self loops.

Example 10.1.2
Two machines, one technician. non-preemptive repairs.
Machine i: time to fail $\sim exp(\lambda_i$, repair $\sim exp(\mu_i)$
Possible states:
- 0: Both machines working
- 1: Machine 1 working on, Machine 2 up
- 2: Machine 2 working on, Machine 1 up

- :(1,2): Machine 1 repairing, 2 broken
- (2,1): 2 under repair, 1 broken

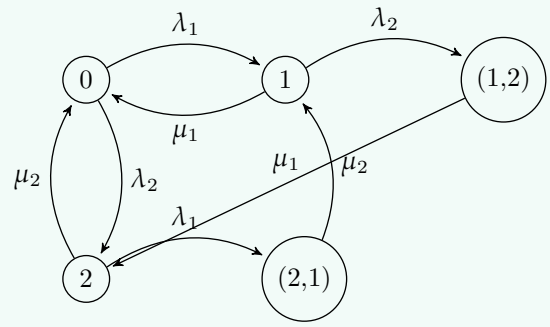


Figure 10.1: Machine Repair CTMC

We can think about the rate/prob that we leave from each state as competiting exponentials.

Question 1: If we start from (1,2), what is the expected time until both machines are working again? (τ is going from state a to state b, T is the time in a state)

$$E[\tau_{(1,2),0}] = \underbrace{E[T_{(1,2)}]}_{\frac{1}{\mu_1}} + E[\tau_{2,0}]$$

$$E[\tau_{2,0}] = \underbrace{\frac{1}{\lambda_1+\mu_2}}_{\text{move out of state}} + \underbrace{\frac{\lambda_1}{\lambda_1+\mu_2}E[\tau_{(2,1),0}]}_{\text{to state (2,1)}} + \underbrace{\frac{\mu_2}{\lambda_1+\mu_2}(0)}_{\text{in state 0}}$$

$$E[\tau_{(2,1),0}] = \frac{1}{\mu_2} + E[\tau_{1,0}]$$

$$E[\tau_{1,0}] = \frac{1}{\mu_1+\lambda_2} + \frac{\lambda_2}{\lambda_2+\mu_1}E[\tau_{(1,2),0}] + \frac{\mu_1}{\lambda_2+\mu_1}(0)$$

Question 2: Start in (1,2), whats the chance that 2 fails again before 1 fails again?
What has to happen for this to occur? We go from state $(1,2) \to 2$, which means only 2 is broken. Then $2 \to 0$, otherwise it wouldn't happen. Finally, 2 has to fail, $0 \to 1$

Example 10.1.3

M/M/1 queue.
Exponential arrival rate (λ), service rate at rate μ, with 1 server.
(see birth death chain for example)

$$E[B] = E[B_1] = E[\tau_{1,0}]$$

$$E[B_1] = \frac{1}{\lambda+\mu} + \frac{\mu}{\lambda+\mu}(0) + \frac{\lambda}{\lambda+\mu}E[B_2]$$

Now we have an infinite number of equations. Thus, we use the same ideas from when we proved null recurrence for a DTMC.